

1 Users' and User accounts

Linux has always pride itself in being a true multi-user operating system. Adding and removing new user accounts is one of the most common task for Linux administrators and it is imperative that he masters this simple and yet essential job. User accounts are also important from the security point of view, as a compromised account with sufficient privileges can make for an easy target.

The `/etc/passwd` file

Whenever you are attempting to logon to any Linux system, your identity is verified by the `/etc/passwd` file. Let us look at its content in a little more detail.

Each line in the file represents one user and contains seven fields separated by colons:

- Login name
- Encrypted password or password placeholder
- UID (user ID) number
- Default GID (group ID) number
- "GECOS" information: full name, office, extension, home phone
- Home directory
- Login shell

Most modern Linux systems don't show the encrypted password now, since with modern hardware they can be quickly decrypted compromising on the security. They are stored typically in a shadow file that isn't readable in text format. This mechanism of hiding even the encrypted password is known as the shadow password mechanism.

For example, the entry for the user `ntp` [Network time protocol] in the `/etc/passwd` file is as follows

```
ntp:x:116:124::/home/ntp:/bin/false
```

We will now look at each of the seven fields in a little more detail.

Login Name

Needs no explanation

Password

The `x` character is displayed when the user's password is stored in `/etc/shadow` in an encrypted format.

User Identification Number or UID

UIDs are 32-bit unsigned integers. It is a recommended practice though, to limit the largest value to a 16-bit range in order to maintain compatibility